Use case 1 Beneficiary payment integrity

Problem and client

Social welfare, scholarships, agricultural assistance and subsidy schemes may suffer duplicate registration, incorrect eligibility, payments beyond an entitlement period or changes in beneficiary bank details without adequate verification. The proposed buyer is the scheme department, with Finance and the state IT department supporting payment and identity integration.

Minimum data and joins

Dataset	Required fields	Link or purpose
Beneficiary master	Scheme beneficiary ID; person or family token; date of birth; location; approved status; validity dates	Resolve the person and the actual entitlement unit
Entitlement and rules	Scheme version; eligibility conditions; benefit amount; frequency; exclusions; effective dates	Reproduce eligibility as of the covered period
Payment history and requests	Payment ID; beneficiary ID; period covered; amount; status; bank token; transaction reference; reversal link	Match scheduled entitlements to actual settlement
Verification and event records	Application evidence; authorised eligibility flags; verified death or status events; event dates; source confidence	Support review; restricted sources require authorised access
Master change logs	Old and new bank tokens; operator ID; timestamp; approval reference	Detect changes and unusual registration behaviour

Where person-level identity is unavailable, use a scheme-specific combination of name, date of birth, address and relationship information for candidate matching. Do not automatically merge records from fuzzy matching. Household and individual entitlements require separate logic.

Detection design

Identity checks Normalise identifiers and names; detect duplicate active records within a scheme. Build candidate links across schemes only where their rules prohibit concurrent benefits. A shared mobile number or bank account is supporting evidence, not proof of duplication.

Eligibility checks Apply versioned conditions to the entitlement period. Authorised employment or income flags may support review where the scheme excludes those categories. Unknown, outdated or contradictory eligibility data goes to a data-correction queue.

Payment checks Reconcile gross scheduled entitlement, credited payments, returns, reversals and retries. Detect repeat settled payments for the same entitled unit and period, excessive cumulative benefits and payments outside the permitted period.

Operator patterns Identify unusually concentrated bank changes or registrations by a user, device or service point where those logs exist. Compare similar service points and exclude approved batch migrations.

Feasible Rajasthan pathway

Jan Aadhaar provides a potential family and member linkage foundation [8]. Scheme MIS, treasury records and approved payment-system extracts can supply transaction information. Public documentation does not establish access to operational datasets or APIs; a state-approved data agreement is required. PFMS validation and rejection documentation illustrates why banking errors should be routed for correction as well as reviewed for payment risk. [9]

Beneficiary rules and operational workflow

Rule	Implementation logic	Review or control
B01 Repeated entitlement	Same canonical entitlement unit + scheme + benefit type + covered period, net settled amount exceeds authorised entitlement. Exclude linked reversals and approved adjustments.	Show each payment and entitlement calculation; verify duplicate obligation.
B02 Conflicting enrolment	High-confidence person match has overlapping active benefits in schemes explicitly marked mutually exclusive for that period.	Check policy exceptions and evidence before any change.
B03 Payment after verified event	Covered entitlement falls after a verified death or other terminating event and after the scheme-specific allowed period.	Exclude legitimate arrears and survivor benefits; seek identity confirmation.
B04 Shared destination cluster	Distinct unrelated entitlement units converge on one bank token, with unusual scale relative to scheme peers and independent corroboration.	Check guardians, joint accounts and approved group arrangements.
B05 Bank-change concentration	Unusual concentration of bank replacements by an operator immediately before payment, no matching approvals.	Verify approvals and beneficiary instructions; investigate operator access.
B06 Eligibility mismatch	Reliable eligibility evidence contradicts a versioned scheme condition as of the applicable assessment date.	Notify scheme reviewer; missing evidence remains a data-quality issue.

End to end process

Ingest records → reconcile totals → resolve candidate identities → calculate entitlements → generate evidence-backed alerts → assign district or scheme reviewer → obtain documents or field confirmation → record a reasoned decision → execute authorised master correction, payment correction or recovery → verify next-cycle payments and close the case.

The reviewer should classify each case as legitimate exception, data error, administrative overpayment, suspected fraud or confirmed outcome. An authorised officer decides on any withholding, recovery or escalation. The system must support correction, notice, grievance and restoration of an incorrectly affected benefit.

Illustrative case

A fictional scholarship beneficiary receives two settled payments of Rs 10,000 for one academic-period entitlement of Rs 10,000. The system links both transaction references and checks for reversals, top-ups and permitted supplementary benefits. If the reviewer confirms duplication, Rs 10,000 is the confirmed overpayment; recovery is recorded only after money is received or a valid adjustment is posted.

Pilot and acceptance

Pilot one cash scheme across three districts, using 12 months of historical transactions and at least one live payment cycle. This is a planning assumption. Initially run in observation mode. A proposed first rule pack comprises B01–B06 with deterministic entitlement checks before optional anomaly scoring.

Acceptance requires complete reconciliation to source totals, adjudication of a stratified alert sample and a random sample of unflagged records, usable evidence screens, and a documented correction and appeal path. Set precision and review-capacity thresholds after baseline assessment. Track verified overpayments, recovery actually credited, repeat errors, review time and wrongful-intervention reversals.

Boundary This module focuses on beneficiary entitlement and disbursement. Provider treatment fraud, ration diversion and physical delivery quality need additional sector data and are not inferred from a beneficiary registry alone.

Use case 4 Payroll and pension integrity

Problem and client

Recurring expenditure can include overlapping salary payments after transfers, payments beyond a service event, incorrect allowances, duplicate arrears and pension calculations that do not match the authorised entitlement. The proposed buyer is Finance, treasury or personnel, supported by departmental HR and pension-sanctioning authorities.

Minimum data and joins

Dataset	Required fields	Link or purpose
HR and posts	Employee ID; authorised person token, office and DDO, post, pay level, employment status, appointment and retirement dates	Establish active service and authorised pay basis
Service events	Transfer; relief; joining, suspension, leave, separation; effective dates; approved orders	Reproduce responsibility and entitlement for a covered period
Payroll and payment	Employee, component, pay period, arrear covered period; gross and net amount; settlement; reversal reference	Detect duplicate pay and compare actual with expected entitlement
Pension authorisation	PPO; pensioner and recipient identity; benefit type; effective dates; rates; commutation; revisions; linked salary adjustments	Keep employee, pensioner and family-pension recipient distinct
Pension settlement and verification	Covered period; settlement; verified death events; life-certificate status; bank changes, approved exceptions	Check recurrence and route identity or verification discrepancies

Detection design

Person and temporal matching Match employee identities across DDOs while preserving service-event dates. Compare the periods to which payments relate, rather than only their posting month. Approved transfer pay, deputation, part-time contracts and arrears may legitimately produce multiple records.

Entitlement recalculation Reconstruct pay and pension components from effective-dated rules, sanctioned orders and permitted adjustments. Identify missing approval, repeated arrears or an allowance that continues after its qualifying condition ends. Keep rounding and retroactive revisions explicit.

Event checks Compare payment entitlement periods with retirement, separation, death or suspension events. Treat a missing or expired life certificate as a verification task; it is not evidence that a recipient has died.

Pension exceptions Re-employment and salary-pension interaction depend on the relevant rules. Multiple authorised pensions and family pensions may be legitimate. Do not flag all concurrent salary and pension as overpayment. Separate the deceased employee from the surviving recipient before mortality matching.

Implementation pathway

Rajasthan IFMS 3.0 public documentation covers employee registration, transfer relief and joining, stop-salary processes, absentee statements and allowance sanctions [11]. These offer potential integration points, subject to authorised access and data completeness. The UK NFI provides an operational precedent for payroll and pension matching followed by investigation [6].

Payroll and pension rules and workflow

Rule	Implementation logic	r control
S01 Overlapping salary	Same high-confidence employee identity receives base pay for overlapping service days from different DDOs without an authorised allocation.	Review transfer, deputation and arrears before calculating duplication.
S02 Post-event entitlement	Salary component covers days after its authorised end, or pension covers a period disallowed after a verified recipient event.	Exclude legitimate final dues, arrears and survivor entitlements.
S03 Excess allowance	Paid component exceeds the effective rule or continues without a valid sanction or qualifying condition.	Recalculate using applicable orders and exception approvals.
S04 Duplicate arrears	Same person + component + covered period + revision order produces overlapping settled arrears beyond approved difference.	Compare period allocations, supplementary bills and reversals.
S05 Pension mismatch	Paid pension differs from effective PPO entitlement after applicable revision, commutation and restoration rules.	Pension authority confirms rate and authorisation history.
S06 Record verification risk	Missing service order, unexplained bank change or overdue verification in an active payment record.	Request evidence; absence of a certificate is not a mortality finding.

End to end process

Connect HR, service orders, payroll, PPOs and settlement → resolve person and recipient identities → construct an effective-dated service and entitlement timeline → calculate expected components → generate alerts → obtain HR or pension-authority explanation → authorise correction, recovery or continued payme[illegible] → reconcile next-cycle results and close.

Payroll staff, HR approvers and treasury reviewers should have separate permissio[illegible] djustment records the authorised order, reason, affected periods and notice status. The system needs an urge[illegible] ction path where an innocent employee or pensioner is affected by stale records.

Illustrative case

A fictional employee transfers midway through a month. Both DDOs pay a full month of basic salary. The service timeline shows the first DDO should cover the first 15 days and the second the balance. The system computes the possible overlap using the applicable pay-period convention, then checks for subsequent reversal and any authorised adjustment. Recovery is considered only after the department confirms the net overpayment.

Pilot and acceptance

Use one department, a selected group of DDOs and 12 months of payroll with effective service orders. Start with duplicate salary and arrears checks; add pension only when PPO and recipient records are linked. Check draft pay bills before the cycle closes and settled payments afterwards. Keep late corrections and supplemental bills in scope.

Acceptance includes transfer cases, split-month entitlement, arrears, deputation, re-employment, commutation restoration and survivor-pension scenarios. Review a random unflagged sample as well as high-risk cases. Track verified overpayment, recovered amount, recurrence, time to master correction and overturned adverse actions.

Positioning This is expenditure assurance, not an employee attendance-surveillance product. Attendance is used only where it changes entitlement under applicable rules and is sufficiently reliable.